

This article walks you through the fundamental concepts you need to learn as you start using Salt.

Installation

Let's begin by installing Salt. In this article, a Debian based distro—Ubuntu 14.04 LTS—is used; for other distros, please check the installation manuals. Salt is designed based on the client-server model. Let us assume a simple infrastructure environment, as shown in Figure 1. The administrator considers his/her machine as the master and other machines as clients. Salt installation happens in two steps. In the first one, we install the *Salt Master* package in the master machine with the following commands as per the official recommendation (<https://docs.saltstack.com/en/latest/topics/installation/ubuntu.html>):

```
$ sudo add-apt-repository ppa:saltstack/salt
$ sudo add-apt-repository ppa:saltstack/salt2015-5
$ sudo apt-get install python-software-properties
$ sudo apt-get install software-properties-common
$ sudo apt-get update
$ sudo apt-get install salt-api
$ sudo apt-get install salt-cloud
$ sudo apt-get install salt-master
$ sudo apt-get install salt-ssh
$ sudo apt-get install salt-syndic
```

In the second step, we install the Salt Minion package in client machines with the following commands, again as per the official recommendation:

```
$ sudo add-apt-repository ppa:saltstack/salt
$ sudo add-apt-repository ppa:saltstack/salt2015-5
$ sudo apt-get install python-software-properties
$ sudo apt-get install software-properties-common
$ sudo apt-get update
$ sudo apt-get install salt-api
$ sudo apt-get install salt-cloud
$ sudo apt-get install salt-minion
```

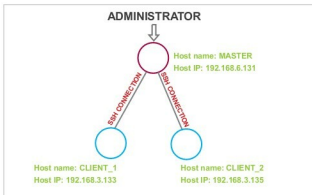


Figure 1: Set-up for admin to install SaltStack components using SSH connections

```
$ sudo apt-get install salt-ssh
$ sudo apt-get install salt-syndic
```

For this step, administrators can use conventional SSH connections as shown in Figure 1 for installation in all the client machines.

Salt configuration

After the installation of Salt Master on the master machine, we need to configure it by modifying the corresponding configuration file called *master* (which is present in the */etc/salt* directory), using any text editor (as shown in Figure 2) to set the master IP address by uncommenting the *interface* file. Before this, it's better to stop the Salt Master daemon running on the master machine with the following command:

```
$ sudo service salt-master stop
```



Figure 2: Salt Master configuration

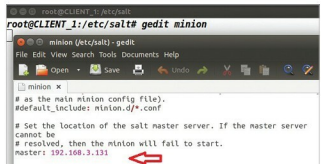


Figure 3: Salt Minion configuration

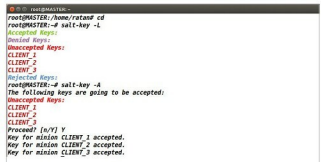


Figure 4: Salt Master-Salt Minions authentication